

Fall 2025.

**Physics 501. Methods of Theoretical Physics I.
Yurii Shylnov.**

Homework # 5.

Due to November 18th, 2025.

Read the corresponding chapters. If you forgot anything necessary from your core math courses, find the information in the textbook, online or whatever works for you. Solve the problems. It should be done in a very detailed and logical way. You need to understand every step of your solution.

Partial Differential Equations in a rectangular system of coordinates.

1. **Mark Pinsky. Section 2.1. Steady solutions.** Why do we need them? Problems: 2.1.3, 2.1.7.
2. **Mark Pinsky. Sections 2.2. Homogeneous boundary conditions.** 2.2.1, 2.2.3.
3. **Mark Pinsky. Sections 2.3. Nonhomogeneous boundary conditions.** 2.3.1, 2.3.5, 2.3.7.
4. **Mark Pinsky. Sections 2.5. Multidimensional equations.** 2.5.5, 2.5.9, 2.5.13.